

L05 — Chatbot I: Foundations

NLP, embeddings, generation, and LLM adaptation

Study summary generated from lecture slides. ETH AI for Digital Characters.

Chatbot basics

- Definition: program simulating human conversation (AI not required)
- Rule-based: deterministic, precise, low risk; poor scalability/flexibility
- Retrieval vs generation: encoder-encoder (dot product) vs encoder-decoder

Text representation

- Tokenization: BPE sub-word tokens
- BoW -> tf-idf -> dense embeddings (Word2Vec, GloVe)
- Encoder models (BERT): bidirectional, good for understanding
- Decoder models (GPT): autoregressive, good for generation

Response generation

- Language model: $P(x_{t+1} | x_1, \dots, x_t)$
- Greedy: always highest prob (repetitive)
- Top-k: sample from k most likely tokens
- Top-p (nucleus): smallest set with cumulative prob $\geq p$
- Temperature: sharpens/flattens distribution
- Beam search: keep top-B partial hypotheses

LLM adaptation

- Fine-tuning: update weights on task data (needs compute + data)
- Prompt engineering: in-context examples, no weight updates
- Instruction tuning (FLAN/T5), MMLU multitask pretraining
- RLHF: reward model + policy optimization (InstructGPT/ChatGPT)
- Prompting: zero-shot, few-shot, chain-of-thought, self-ask, ReAct

Optimizations

- PEFT / LoRA: train low-rank adapters, freeze most weights
- Pruning: remove least important weights
- Knowledge distillation: teacher -> smaller student
- Quantization: lower precision (AWQ, SmoothQuant)